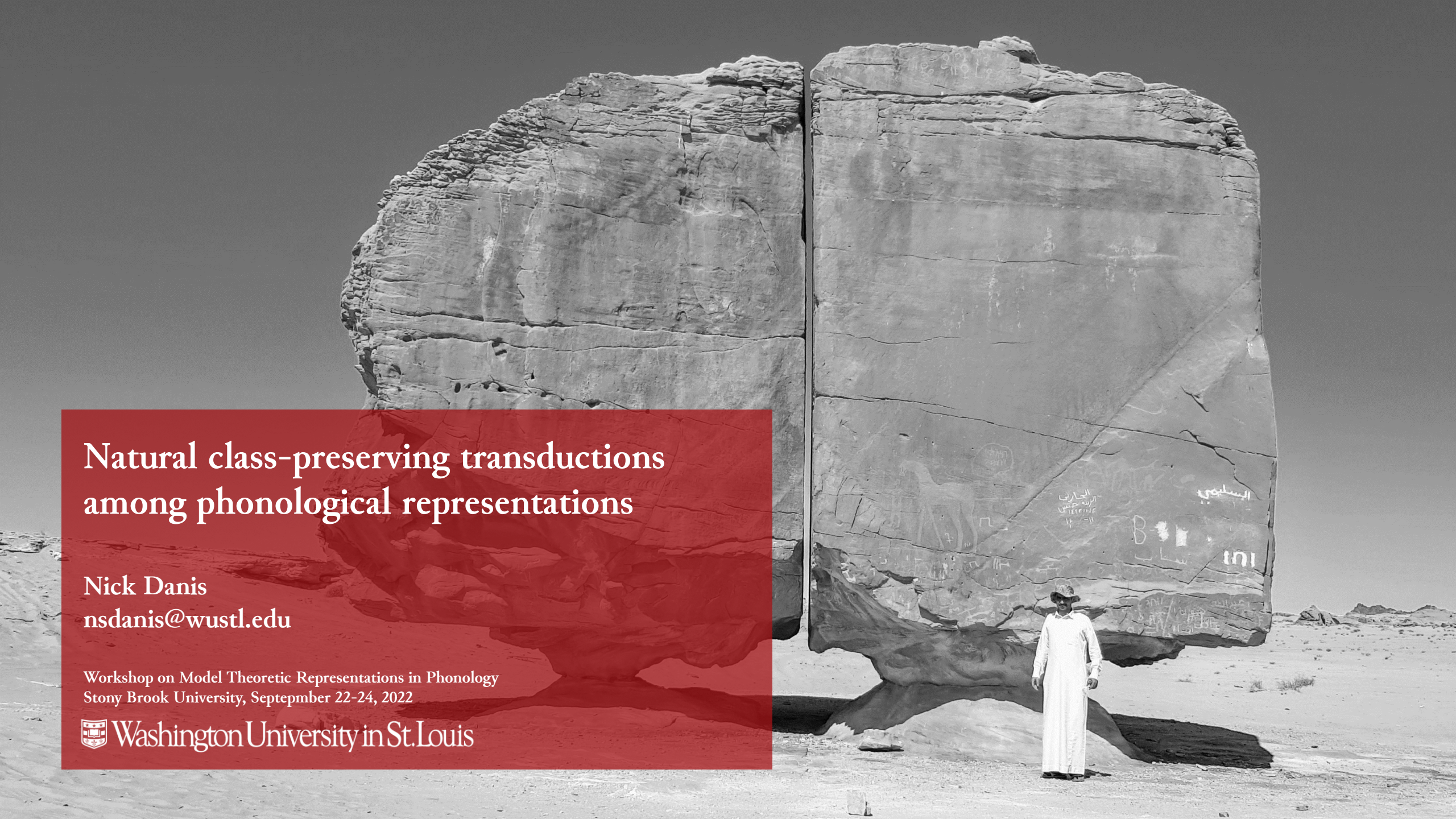


A large, ancient stone tablet, likely the Rosetta Stone, stands in a desert landscape. The tablet is covered in hieroglyphs and is being held by a person in a white robe and hat, providing a sense of scale. The background is a clear, bright sky.

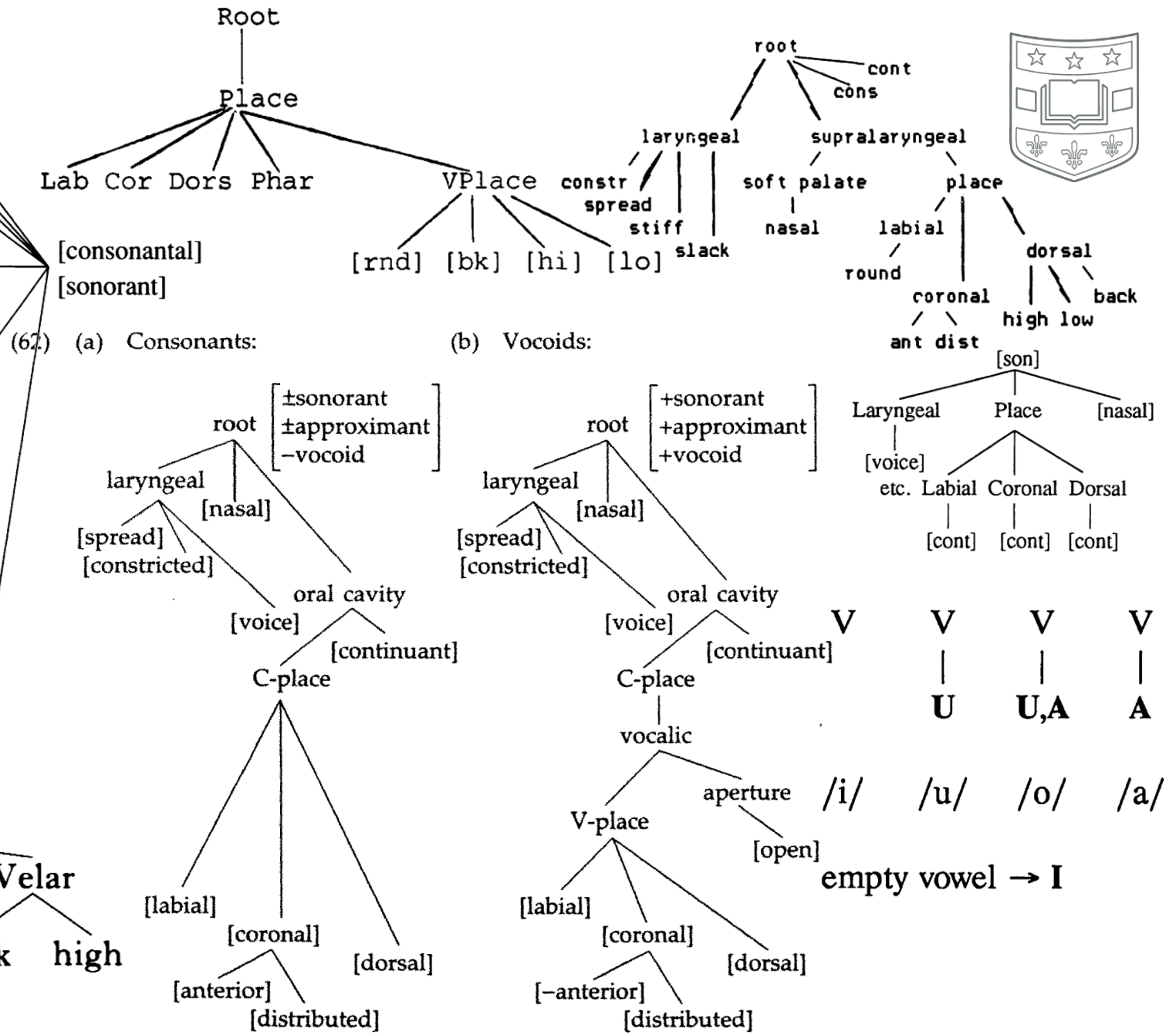
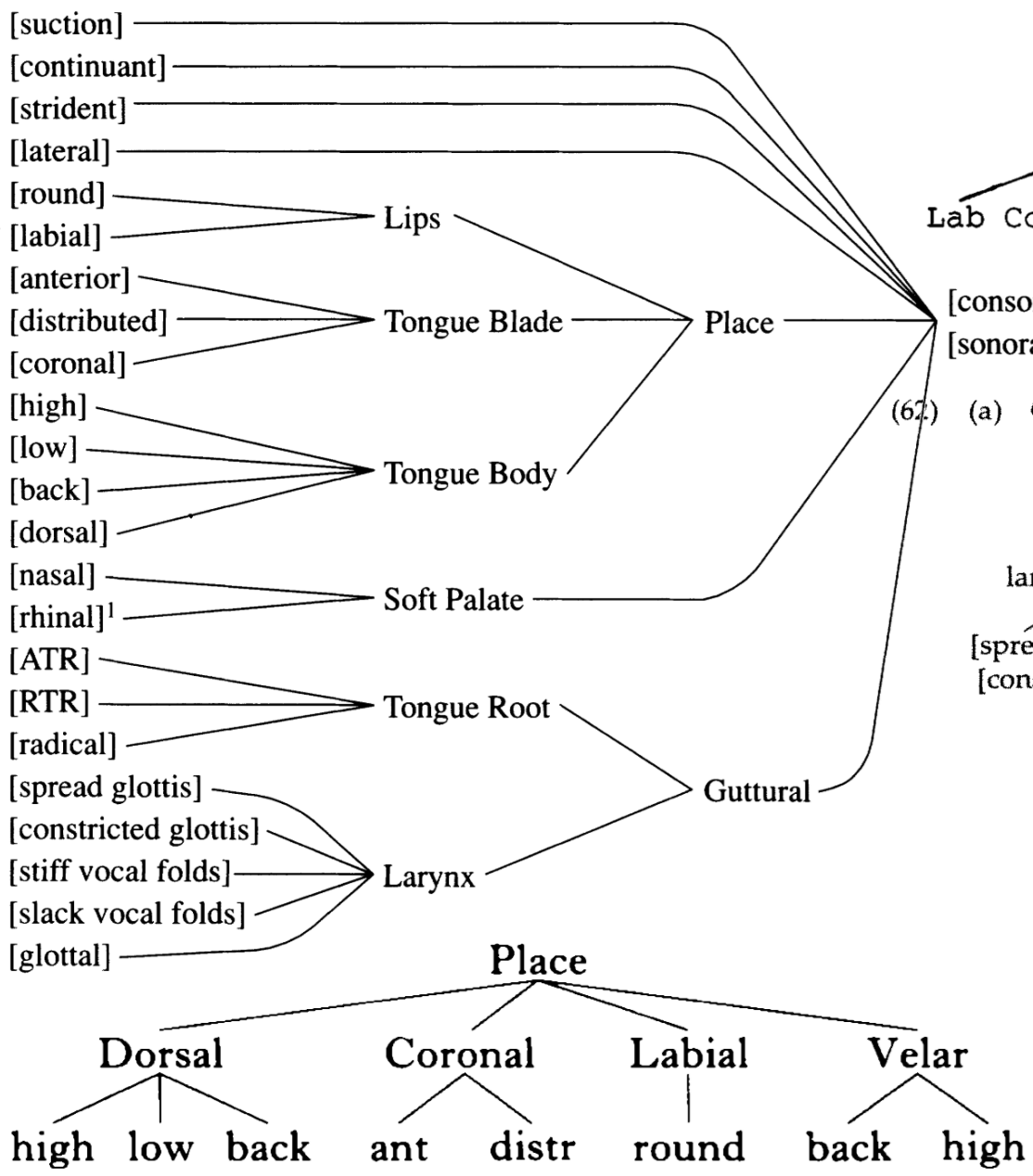
Natural class-preserving transductions among phonological representations

Nick Danis
nsdanis@wustl.edu

Workshop on Model Theoretic Representations in Phonology
Stony Brook University, September 22-24, 2022

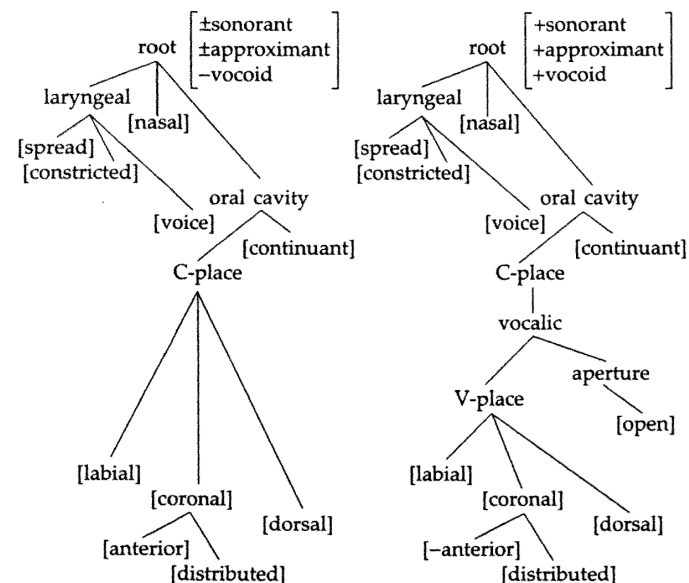


Washington University in St. Louis

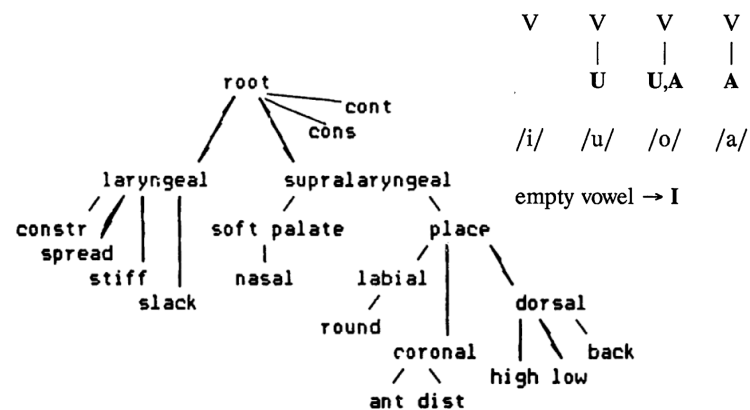
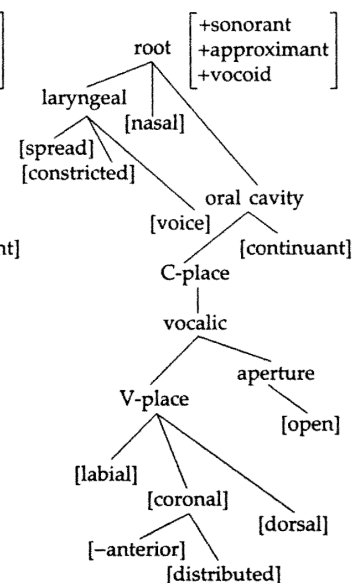


consonants and vowels probably should be natural classes

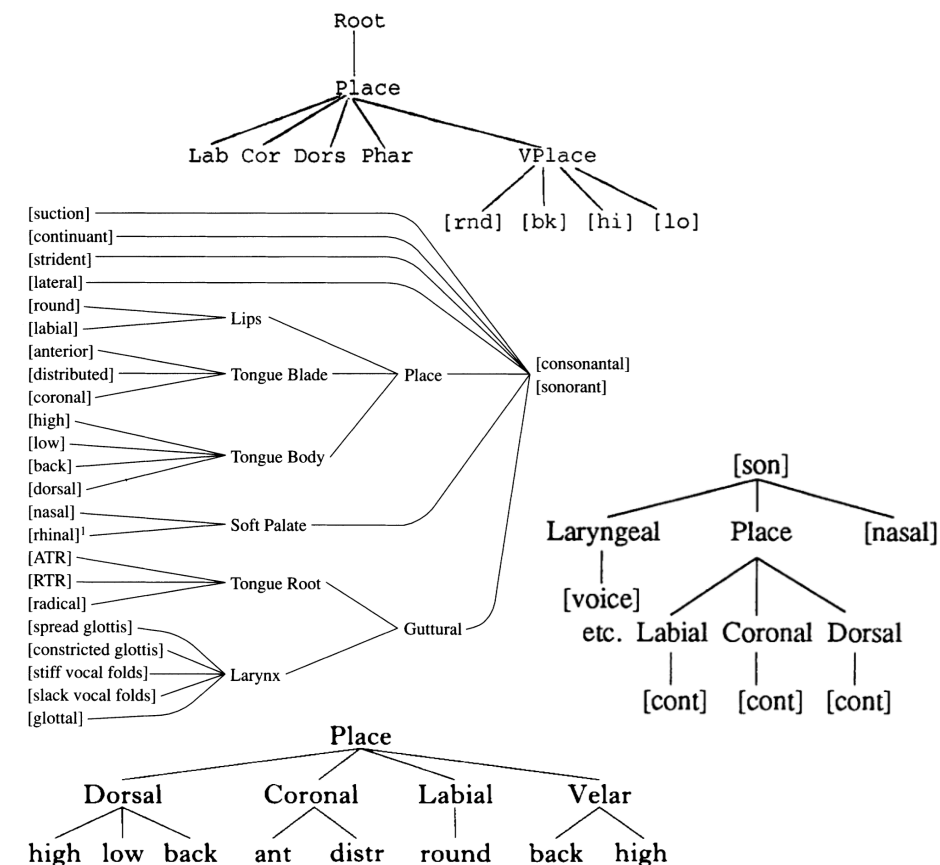
(62) (a) Consonants:



(b) Vocoids:



consonants and vowels probably *shouldn't* be natural classes



big questions



- how can we formally compare phonological representations?
- what can we learn from these comparisons?
- what does it mean for us as linguists?
- what does *anything* mean?

medium answers



- two theories can be shown to be **formally equivalent** using logic and model theory
 - given two representations A and B, a **transduction** between A and B means that any linguistic rule given with structure A can be translated into structure B, and vice versa
 - Strother-Garcia (2019), Danis & Jardine (2019), Oakden (2020), a.o.

Figure 4.5: $\mathcal{M}_{plenty}^{flat}$

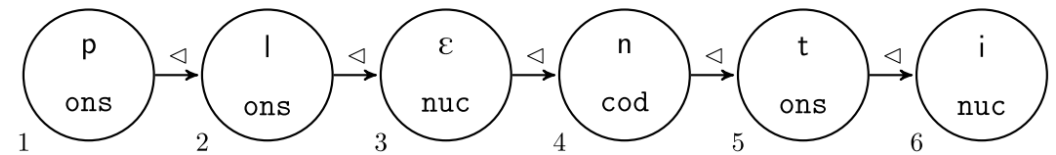
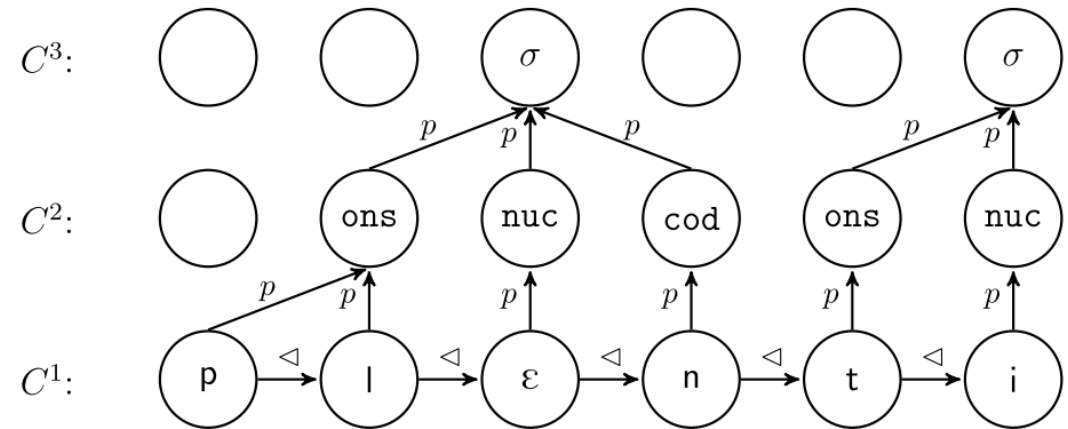


Figure 4.16: $\Gamma_{ft}(\mathcal{M}_{plenty}^{flat})$ fully specified

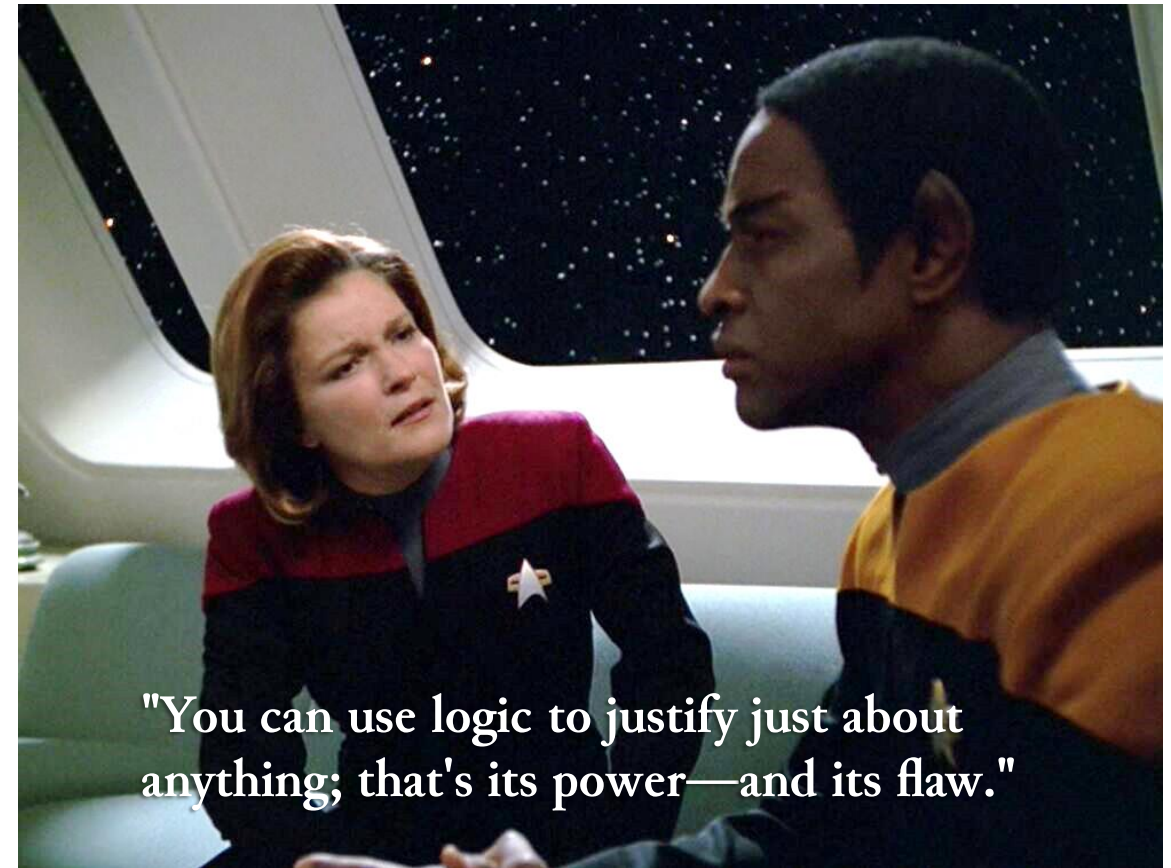


(Strother-Garcia 2019)



medium answers

- not every transduction preserves ideas of *linguistic* equivalence
 - process should respect natural classes, which may be lost in certain transductions
- the property of a **natural-class preserving transduction** is defined to find those logically equivalent representations that also share linguistic intuitions



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natural-class preserving transductions

- A transduction between two representations A and B is **natural-class preserving** iff the set of all natural class extensions of A exactly match those of B
 - A **natural class extension** is a maximal set of segments that share some common structural property

extension: { i u }

shared structure: [+high]

i u
e o
 a

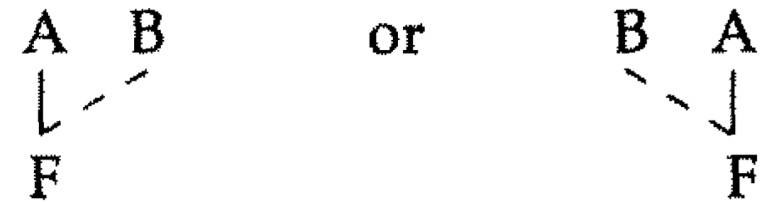
every segment in this set is [+high]
no other segment in the theory is [+high]



assimilation: sharing is caring

- assimilation operates over like things
 - Trubetzkoy (1969), Chomsky and Halle (1968), Hyman (1974), Hayes (1986), Clements & Hume (1995) a.o.

$$X \rightarrow [\alpha F] / [\alpha F]_Y$$



Naturalness of Assimilation

the targets and triggers of an assimilation process should constitute a natural class extension

assimilation: sharing is caring



- "Nevertheless, there is empirical evidence in favor of imposing a limitation on the use of variables with different features in different segments. **The great majority of examples involve only a single feature, and in other cases there clearly seems to be some intrinsic connection between the features involved in the process of assimilation.** At the present juncture, however, we are in no position to formulate these restrictions." (SPE 352)



assimilation: sharing is caring

- Clements & Hume (1995):
 - "Phonological rules perform single operations only." (p. 250)
 - "In the present model, in contrast, assimilation rules are characterized as the association (or "**spreading**") of a **feature or node F** of segment A to a neighboring segment B..." (p. 258)
- If assimilation is the result of spreading, then it directly follows from this that the resulting segments will have shared structure and therefore constitute a nontrivial natural class

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comparing theories

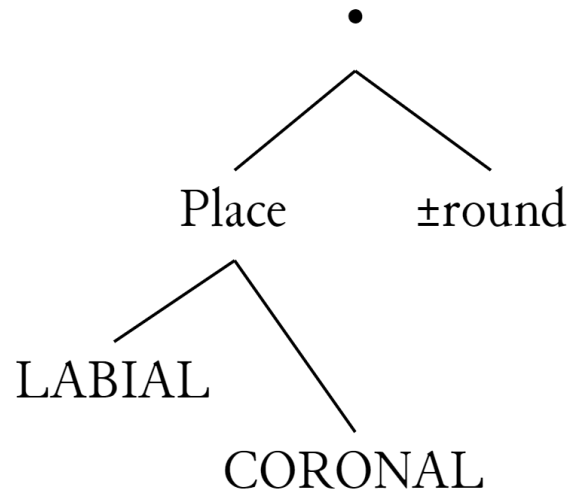
unified place theory

- consonants and vowels share representational primitives
 - e.g. LABIAL C-place, LABIAL V-place
- Sagey (1986), Clements & Hume (1995), a.o.

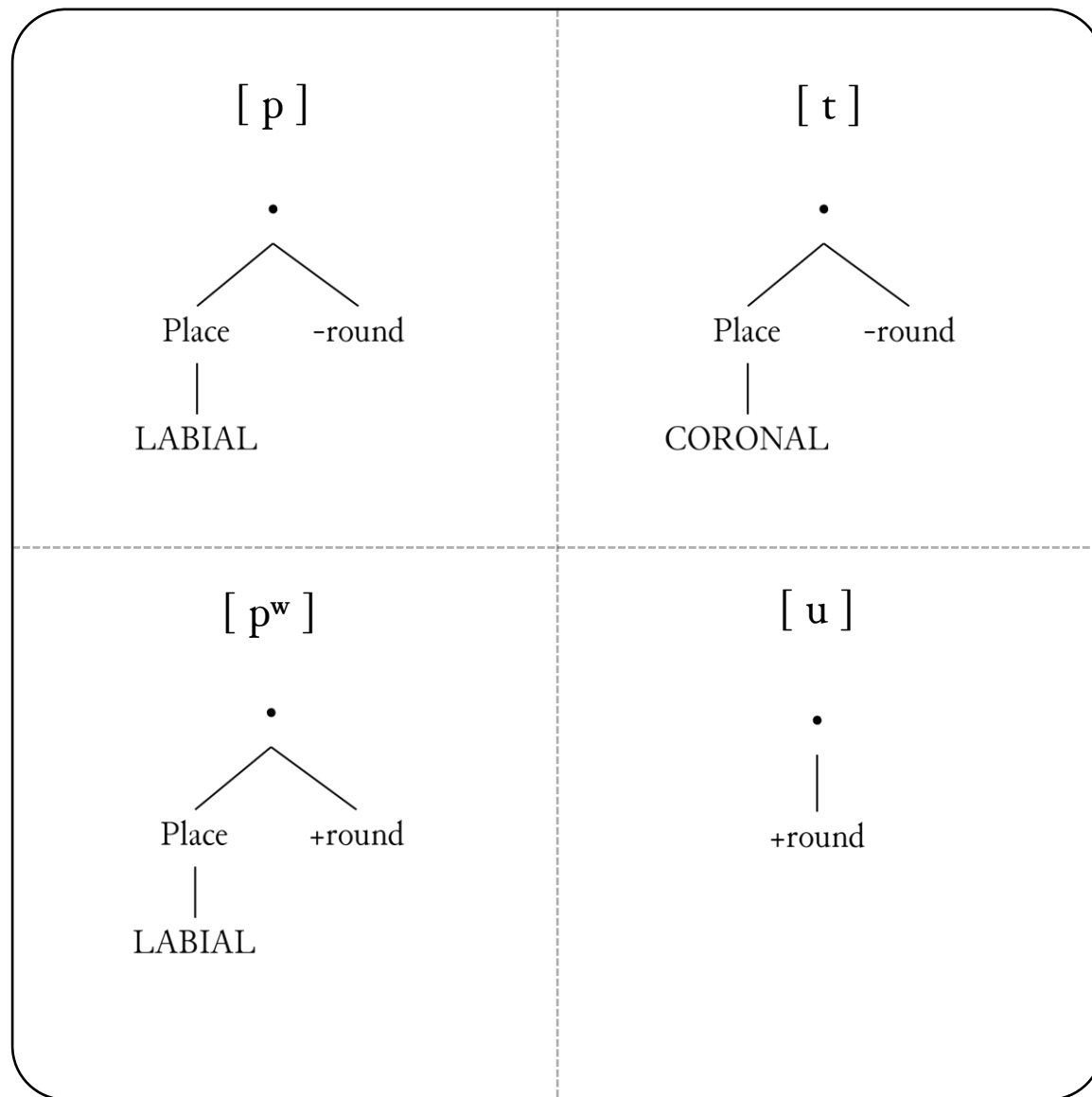
vowel feature theory

- vowel place is largely defined by primitives not used to describe consonant place
 - e.g. [+back], [-round]
- Odden (1991), Ni Chiosain & Padgett (1993), Halle et al. (2000), a.o.

v-feature theory



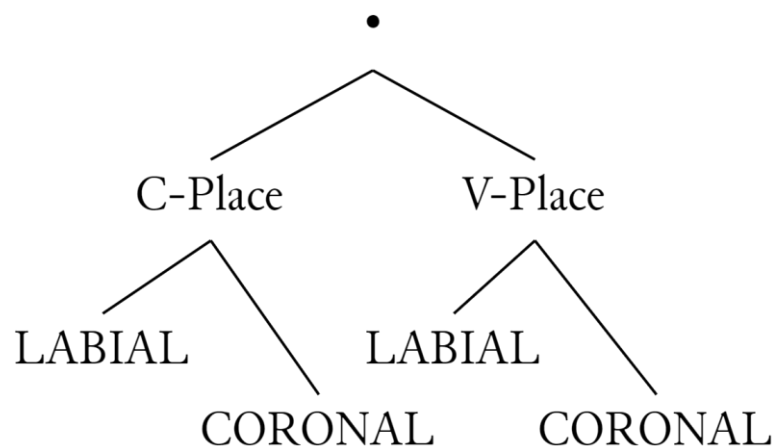
full structure



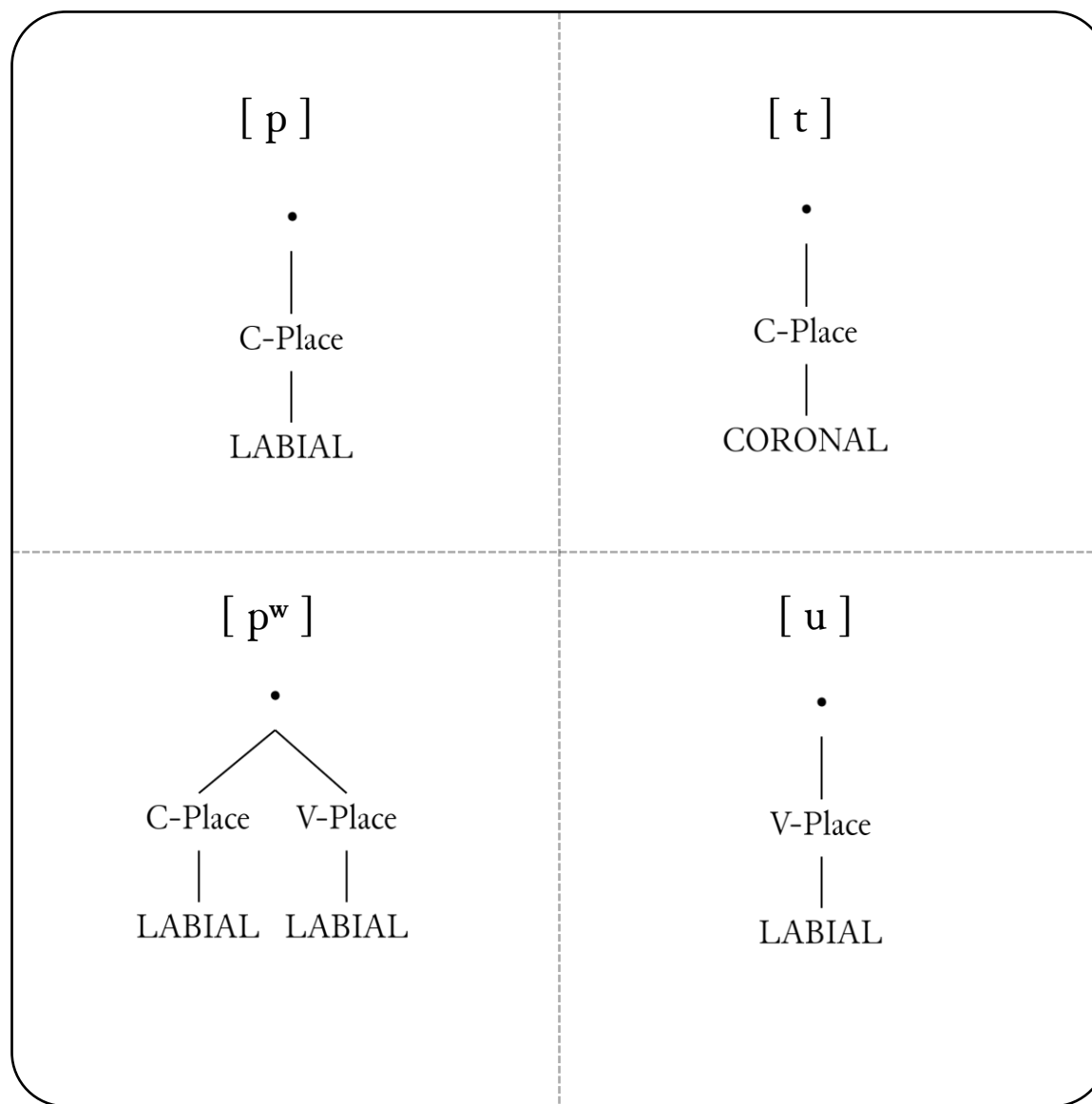
segments



unified place theory



full structure



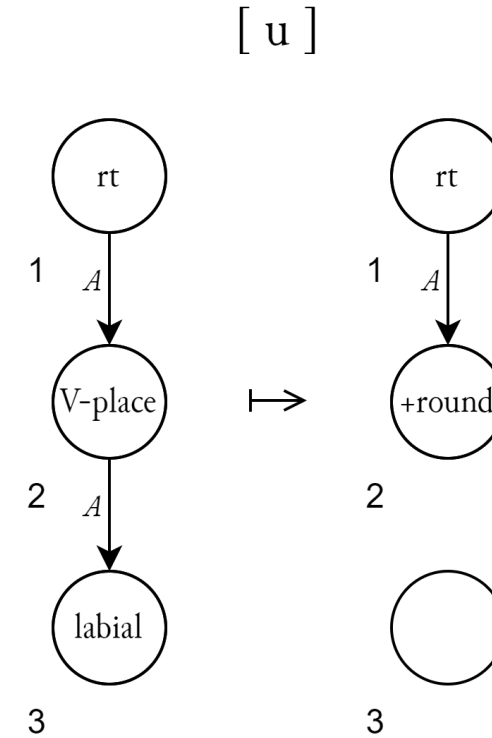
segments





the transduction: unified $\rightarrow$ v-feature

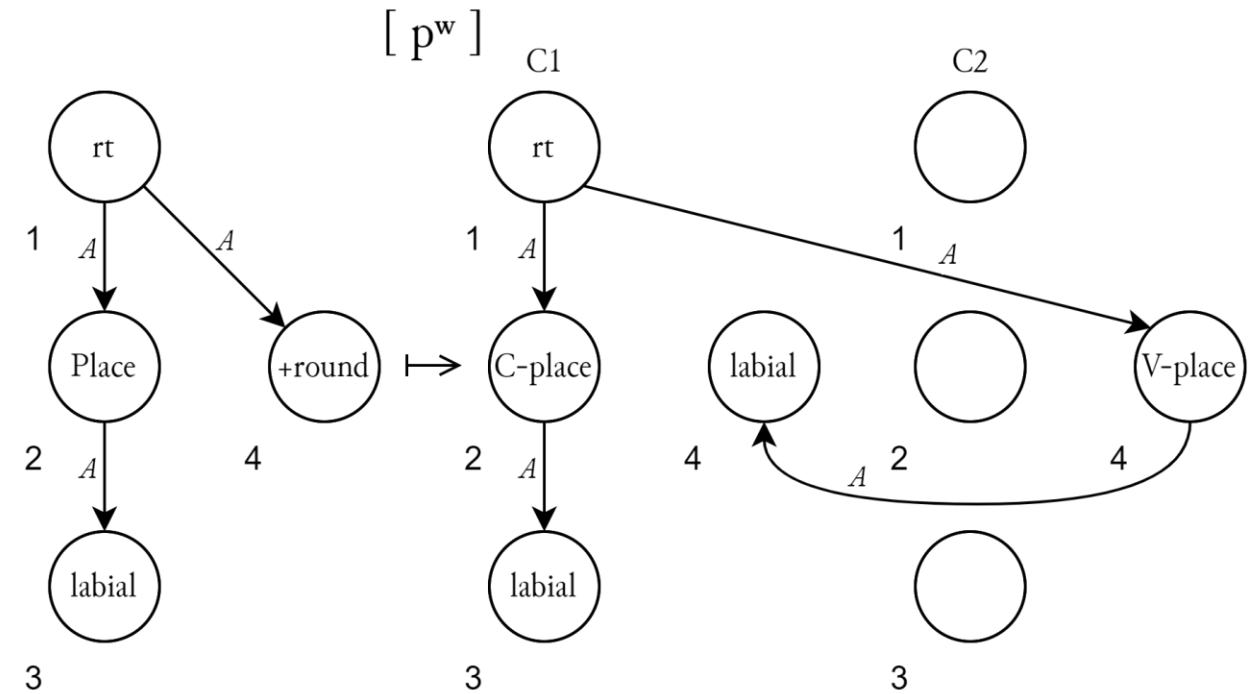
$rt(x^1) = rt(x)$
 $Place(x^1) = C-place(x)$
 $labial(x^1) = labial(x) \wedge C-place(y) \wedge A(y, x)$
 $coronal(x^1) = coronal(x) \wedge C-place(y) \wedge A(y, x)$
 $+round(x^1) = labial(x) \wedge V-place(y) \wedge A(y, x)$
 $-round(x^1) = \neg(labial(x) \wedge V-place(y) \wedge A(y, x))$
 $+front(x^1) = coronal(x) \wedge V-place(y) \wedge A(y, x)$
 $-front(x^1) = \neg(coronal(x) \wedge V-place(y) \wedge A(y, x))$





the transduction: v-feature $\rightarrow$ unified

$rt(x^1) = rt(x)$
 $labial(x^1) = +round(x) \vee labial(x)$
 $coronal(x^1) = +front(x) \vee coronal(x)$
 $C-place(x^1) = Place(x)$
 $V-place(x^2) = +round(x) \vee +front(x)$
 $A(x^2, y^1) = (+round(x) \vee +front(x)) \wedge x = y$
 $A(x^1, y^2) = rt(x) \wedge (+round(y) \vee +front(y))$
 $A(x^1, y^1) = A(x, y) \wedge \neg(+round(y) \vee +front(y))$





unified and v-features are QF-bi-interpretable

and are therefore notational variants?

"A QF transduction is extremely restricted in the degree to which the output can differ from the input because QF is a weak logical language limited to local operations. QF-bi-interpretability can therefore be considered an indication of notational equivalence."

(Strother-Garcia 2019: 39)



enumerating natural class extensions

general procedure:

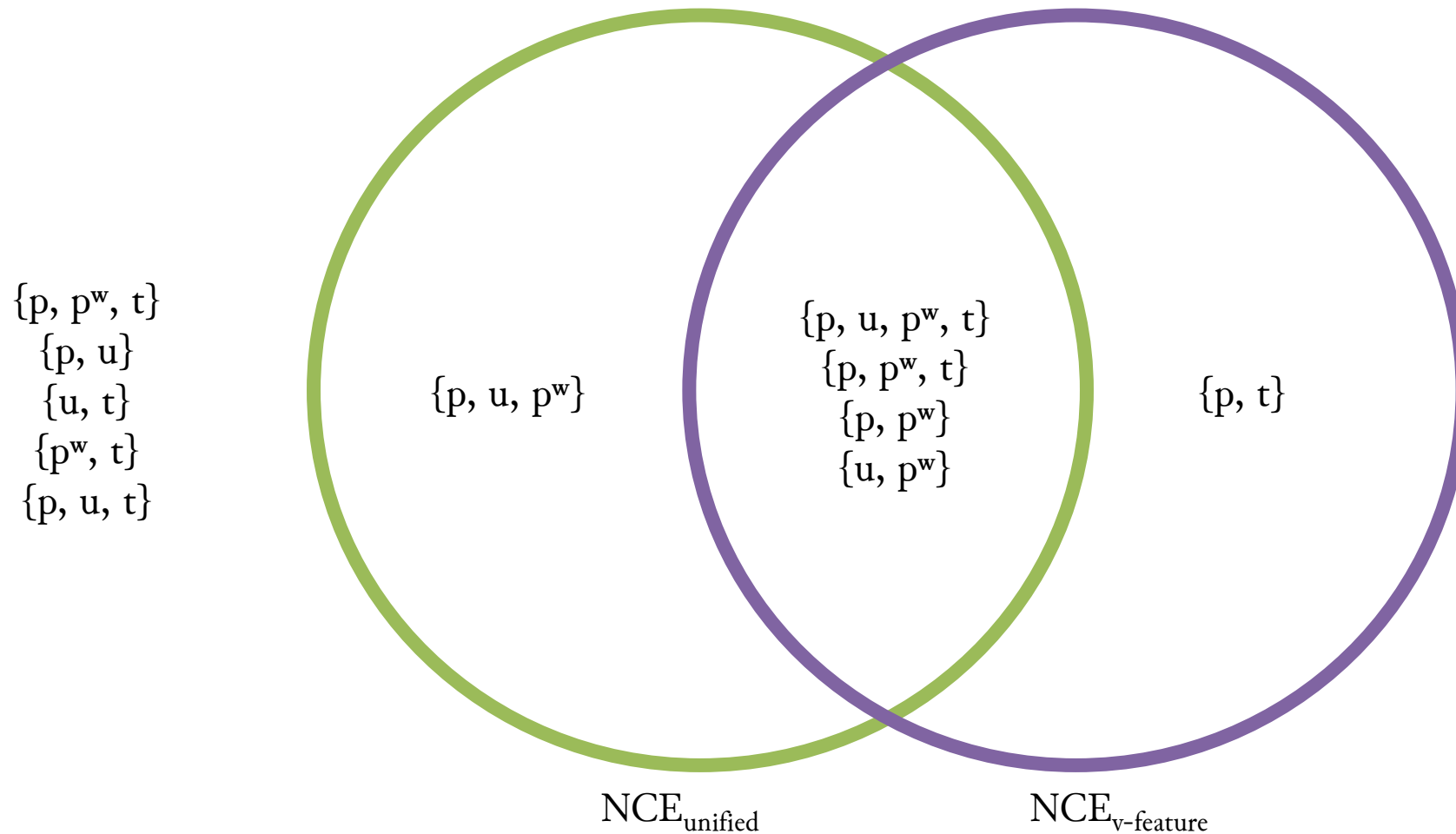
- given the set of possible segments S , find all subsets of S
- for each subset N , find the shared structure G
 - for each segment not in N , check if it also contains G
 - if true, N is not a natural class extension
 - if false, continue
 - N is a natural class extension

enumerating natural class extensions



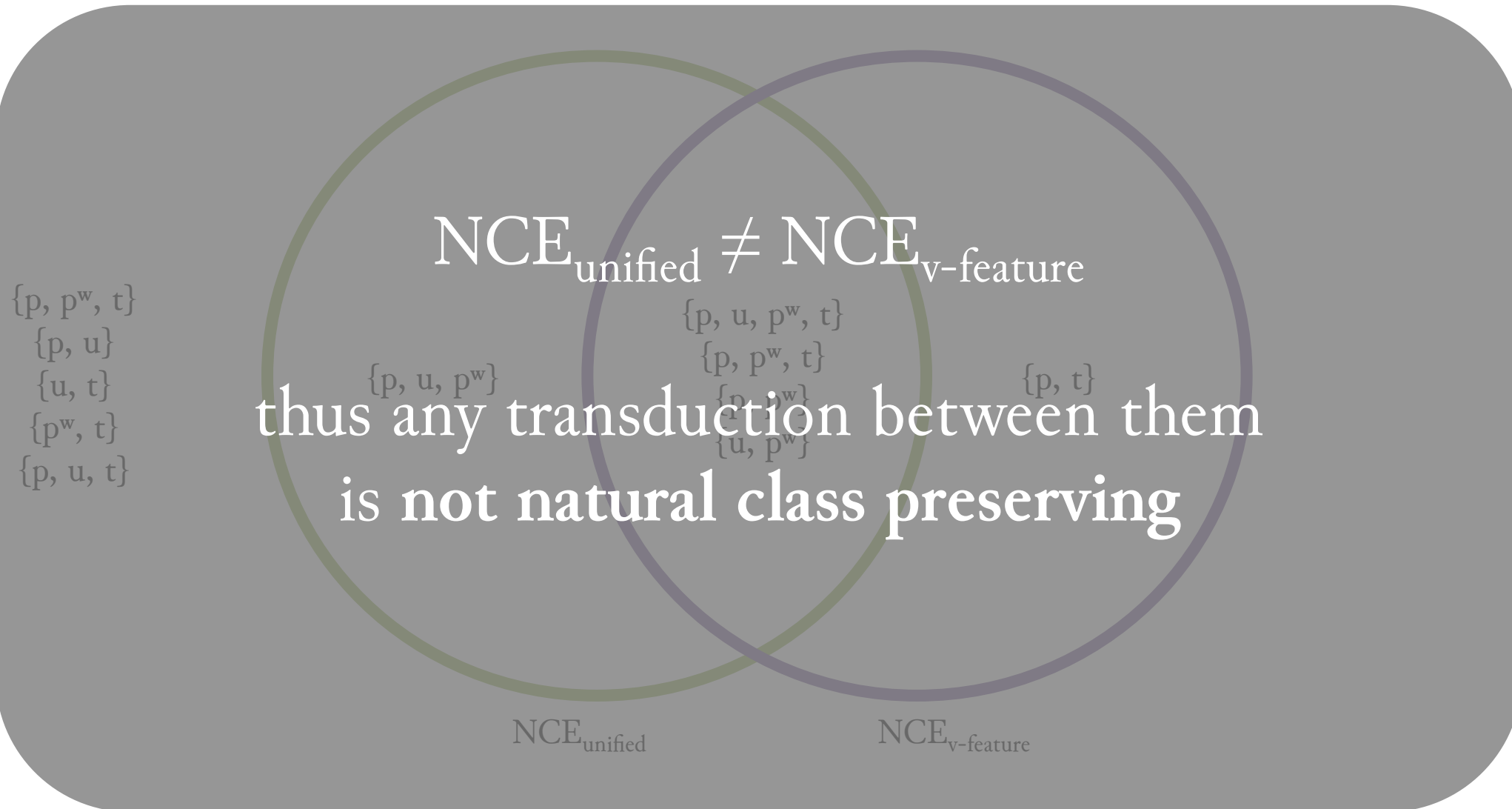
	subset	$\{p, u, p^w, t\}$	$\{p, u, p^w\}$	$\{p, p^w, t\}$	$\{p, p^w\}$	$\{p, t\}$	$\{u, p^w\}$
largest unique shared structure	unified	•	• LABIAL	• C-Place	• C-Place LABIAL	∅	• V-Place LABIAL
	v-feature	•	∅	• Place	• Place LABIAL	• / \ Place -round	• +round

comparing natural class extensions





comparing natural class extensions

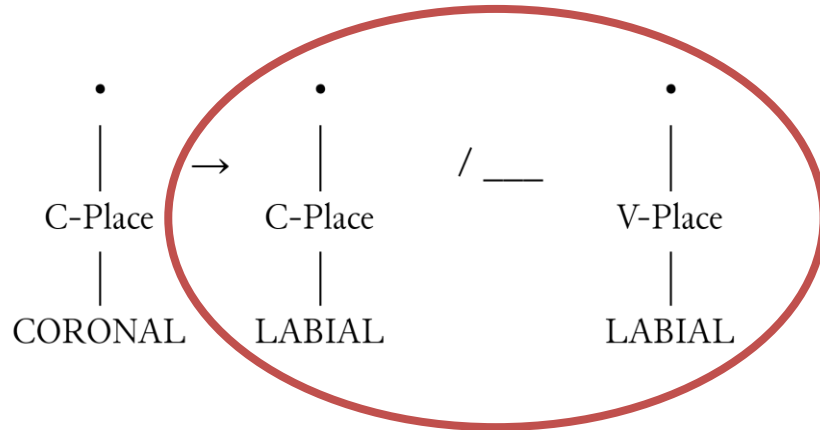


hypothetical assimilation

- $t \rightarrow p / _ u$

Naturalness of Assimilation

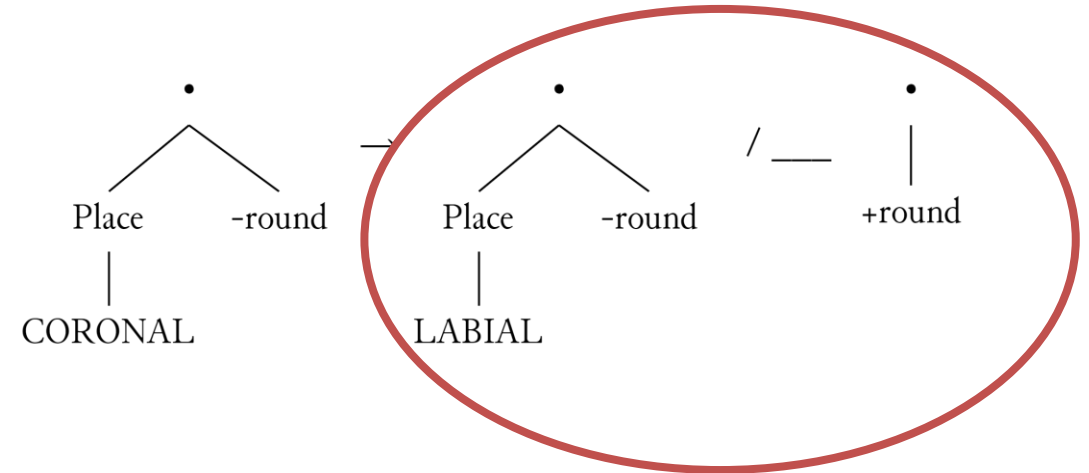
the targets and triggers of an assimilation process should constitute a natural class extension



natural class extension: $\{p, u, p^w\}$

shared structure: LABIAL

- predicts $[p^w]$ also a trigger
- seems reasonable



natural class extension: $\{t, p, u, p^w\}$

shared structure: •

- predicts $[p^w t]$ also triggers
- seems wrong
- not even an assimilation rule

comparing theories



(1) *Conditions for notational equivalence*

- a. Two models do not differ in their empirical predictions.
- b. Two models represent the same set of abstract properties, differing only superficially.

(from Oakden 2021, summarizing Fromkin 2010)

- if we take seriously assumptions like Naturalness of Assimilation  (in whatever formulation), **then a QF-bi-interpretable contrast-preserving transduction is not enough to satisfy (1a) above**

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-
- PRINCETON UNIVERSITY

 Washington University in St. Louis[illegible]

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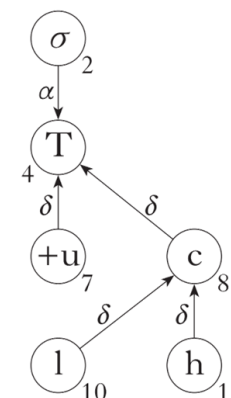
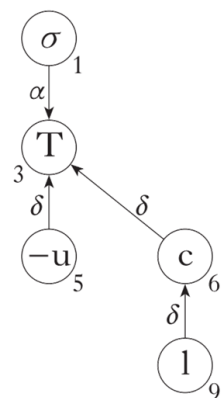
Oakden (2021)



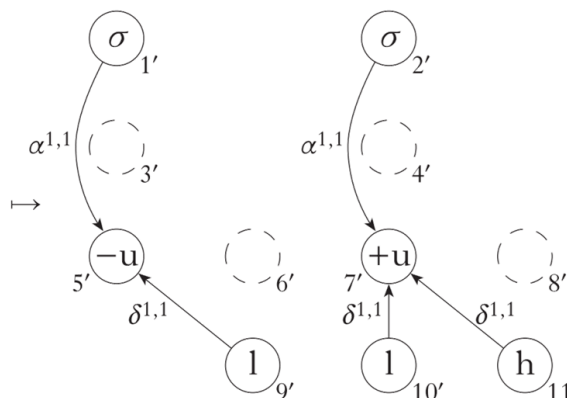
tone	Yip (1989)	Bao (1990)
low level L	$\begin{array}{c} \sigma \\ \\ -u \\ \\ l \end{array}$	$\begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ -u \quad c \\ \quad \\ l \quad l \end{array}$
high level H	$\begin{array}{c} \sigma \\ \\ +u \\ \\ h \end{array}$	$\begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ +u \quad c \\ \quad \\ h \quad h \end{array}$
mid level M	$\begin{array}{c} \sigma \\ \\ -u \\ \\ h \end{array} \text{ or } \begin{array}{c} \sigma \\ \\ +u \\ \\ l \end{array}$	$\begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ -u \quad c \\ \quad \\ h \quad l \end{array} \text{ or } \begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ +u \quad c \\ \quad \\ l \quad l \end{array}$
high falling HM	$\begin{array}{c} \sigma \\ \\ +u \\ / \backslash \\ h \quad l \end{array}$	$\begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ +u \quad c \\ / \backslash \quad \\ h \quad l \quad l \end{array}$

tone	Yip (1989)	Bao (1990)
high rising MH	$\begin{array}{c} \sigma \\ \\ +u \\ / \backslash \\ l \quad h \end{array}$	$\begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ +u \quad c \\ / \backslash \quad \\ l \quad h \quad h \end{array}$
low falling ML	$\begin{array}{c} \sigma \\ \\ -u \\ / \backslash \\ h \quad l \end{array}$	$\begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ -u \quad c \\ / \backslash \quad \\ h \quad l \quad l \end{array}$
low rising LM	$\begin{array}{c} \sigma \\ \\ -u \\ / \backslash \\ l \quad h \end{array}$	$\begin{array}{c} \sigma \\ \\ T \\ / \backslash \\ -u \quad c \\ \quad \\ l \quad h \end{array}$

(38) *separated*



bundled



- Oakden (2021) provides a non-size-preserving QF transduction (above) between two theories of tone sandhi (left), arguing for notational equivalence
- **is this transduction also natural class preserving?**

Table I

Level and contour tonal contrasts in Yip (1989) and Bao (1990).

Oakden (2021)



- yes
- because Python

```
[8]: 1 yip = {
      2     'L' : {'s', '-u'}, ('-u', 'l'),
      3     'H' : {'s', '+u'}, ('+u', 'h'),
      4     'M1' : {'s', '-u'}, ('-u', 'h'),
      5     'M2' : {'s', '+u'}, ('+u', 'l'),
      6     'HM' : {'s', '+u'}, ('+u', 'h'), ('+u', 'l'),
      7     'MH' : {'s', '+u'}, ('+u', 'l'), ('+u', 'h'),
      8     'ML' : {'s', '-u'}, ('-u', 'l'), ('-u', 'h'),
      9     'LM' : {'s', '-u'}, ('-u', 'l'), ('-u', 'h')}
     10 }
     11
     12 bao = {
     13     'L' : {'s', 'T'}, ('T', '-u'), ('T', 'c'), ('c', 'l'),
     14     'H' : {'s', 'T'}, ('T', '+u'), ('T', 'c'), ('c', 'h'),
     15     'M1' : {'s', 'T'}, ('T', '-u'), ('T', 'c'), ('c', 'h'),
     16     'M2' : {'s', 'T'}, ('T', '+u'), ('T', 'c'), ('c', 'l'),
     17     'HM' : {'s', 'T'}, ('T', '+u'), ('T', 'c'), ('c', 'h'), ('c', 'l'),
     18     'MH' : {'s', 'T'}, ('T', '+u'), ('T', 'c'), ('c', 'h'), ('c', 'l'),
     19     'ML' : {'s', 'T'}, ('T', '-u'), ('T', 'c'), ('c', 'h'), ('c', 'l'),
     20     'LM' : {'s', 'T'}, ('T', '-u'), ('T', 'c'), ('c', 'h'), ('c', 'l'),
     21 }
     22
     23 compare_theories(yip, bao)
```

Natural classes unique to theory 1:

Natural classes unique to theory 2:

Natural classes in common:

('LM', 'M1', 'ML')

('LM', 'ML')

('HM', 'M2', 'MH')

('H', 'HM', 'M2', 'MH')

('L', 'LM', 'M1', 'ML')

('H', 'HM', 'LM', 'M1', 'MH', 'ML')

('H', 'HM', 'L', 'LM', 'M1', 'M2', 'MH', 'ML')

('HM', 'L', 'LM', 'M2', 'MH', 'ML')

('H', 'HM', 'MH')

('L', 'LM', 'ML')

('HM', 'MH')

('HM', 'LM', 'MH', 'ML')

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summary

- the use of features/subsegmental structure sorts segments into natural classes based on shared structure
- while transductions can translate from model A to model B, the predicted natural classes present in the system as a whole may still differ in a linguistically significant way
- the definition of natural class preserving transductions is a start to identify comparisons of theories which still involve differing linguistic predictions

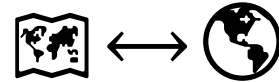
going forward



- the definition of *natural class preserving* is based on the representation themselves—can this property be identified by investigating the transduction rules alone?
- how else can transductions themselves be compared and evaluated from a linguistic standpoint?
- how strongly should our metatheoretical assumptions and expectations about linguistic processes be formalized?



thank you



& thank you Adam Jardine (for helping with the transductions like 3 years ago)
and the audience at the Wash U Linguistics Brown Bag on Sept. 16